

Web Application Development

HTML Basics

LECTURE 2 and 3

Ammar Ahmad Khan

HTML

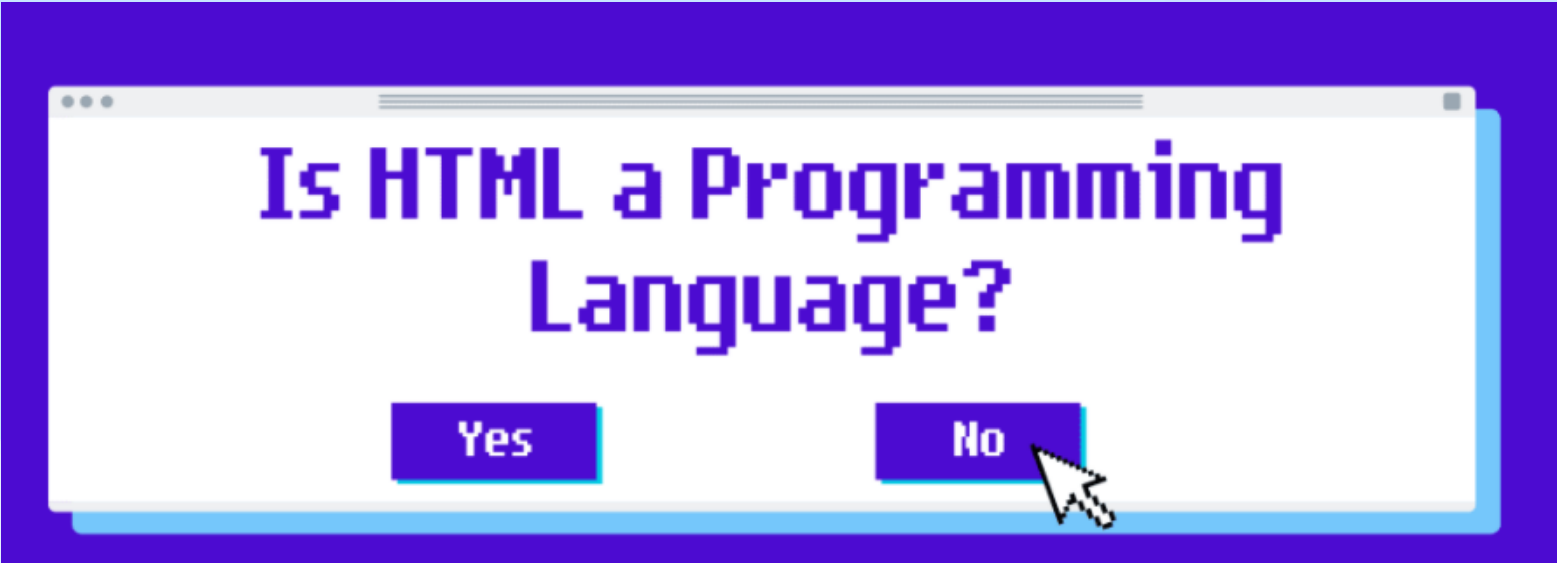
- HTML
 - Stands for HyperText Markup Language
 - HyperText
 - The text that has reference to another text document
 - Markup language
 - Language that is both machine and human readable
- HTML Page
 - HTML pages are text documents containing HTML structure
 - HTML **describes the structure** of a page
 - HTML files are text files with extension “**htm**” or “**html**”

HTML History

- Tim Berners-Lee invented html in 1991

Version	Year
HTML	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML5	2014





Is HTML a Programming
Language?

Yes

No

Text + HTML + CSS = Web Page



HTML <!DOCTYPE> Declaration

```
<!DOCTYPE html>
```

```
<html>
```

```
...
```

```
...
```

```
...
```

```
...
```

```
...
```

```
...
```

```
</html>
```

<!DOCTYPE> Definition and Use

- The <!DOCTYPE> declaration must be the very first thing in your HTML document, before the <html> tag.
- The <!DOCTYPE> declaration is not an HTML tag; it is an instruction to the web browser about what version of HTML the page is written in.
- In HTML 4.01, the <!DOCTYPE> declaration refers to a DTD (Document Type Definition)
- The DTD specifies the rules for the markup language, so that the browsers render the content correctly.
- HTML5 does not require a reference to a DTD.
- There are three different <!DOCTYPE> declarations in HTML 4.01.

Version	Year
HTML	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML 5	2014

<!DOCTYPE> Usage

- Always add the <!DOCTYPE> declaration to your HTML documents, so that the browser knows what type of document to expect.
- The <!DOCTYPE> declaration is NOT case sensitive

Common DOCTYPE Declarations

- **HTML 5**
 - `<!DOCTYPE html>`
- **HTML 4.01**
 - `<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">`
- **XHTML 1.1**
 - `<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">`

Example: HTML Structure

```
<html>
  <body>
    <h1>This is the Main Heading</h1>
    <p>This text might be an introduction to the rest of
      the page. And if the page is a long one it might
      be split up into several sub-headings.</p>
    <h2>This is a Sub-Heading</h2>
    <p>Many long articles have sub-headings so to help
      you follow the structure of what is being written.
      There may even be sub-sub-headings (or lower-level
      headings).</p>
    <h2>Another Sub-Heading</h2>
    <p>Here you can see another sub-heading.</p>
  </body>
</html>
```

Example: HTML Structure

```
<html>
  <body>
    <h1>This is
    <p>This text
      the page.
      be split
    <h2>This is
    <p>Many long
      you follo
      There may
      headings)
    <h2>Another
    <p>Here you
  </body>
</html>
```

This is the Main Heading

This text might be an introduction to the rest of the page. And if the page is a long one it might be split up into several sub-headings.

This is a Sub-Heading

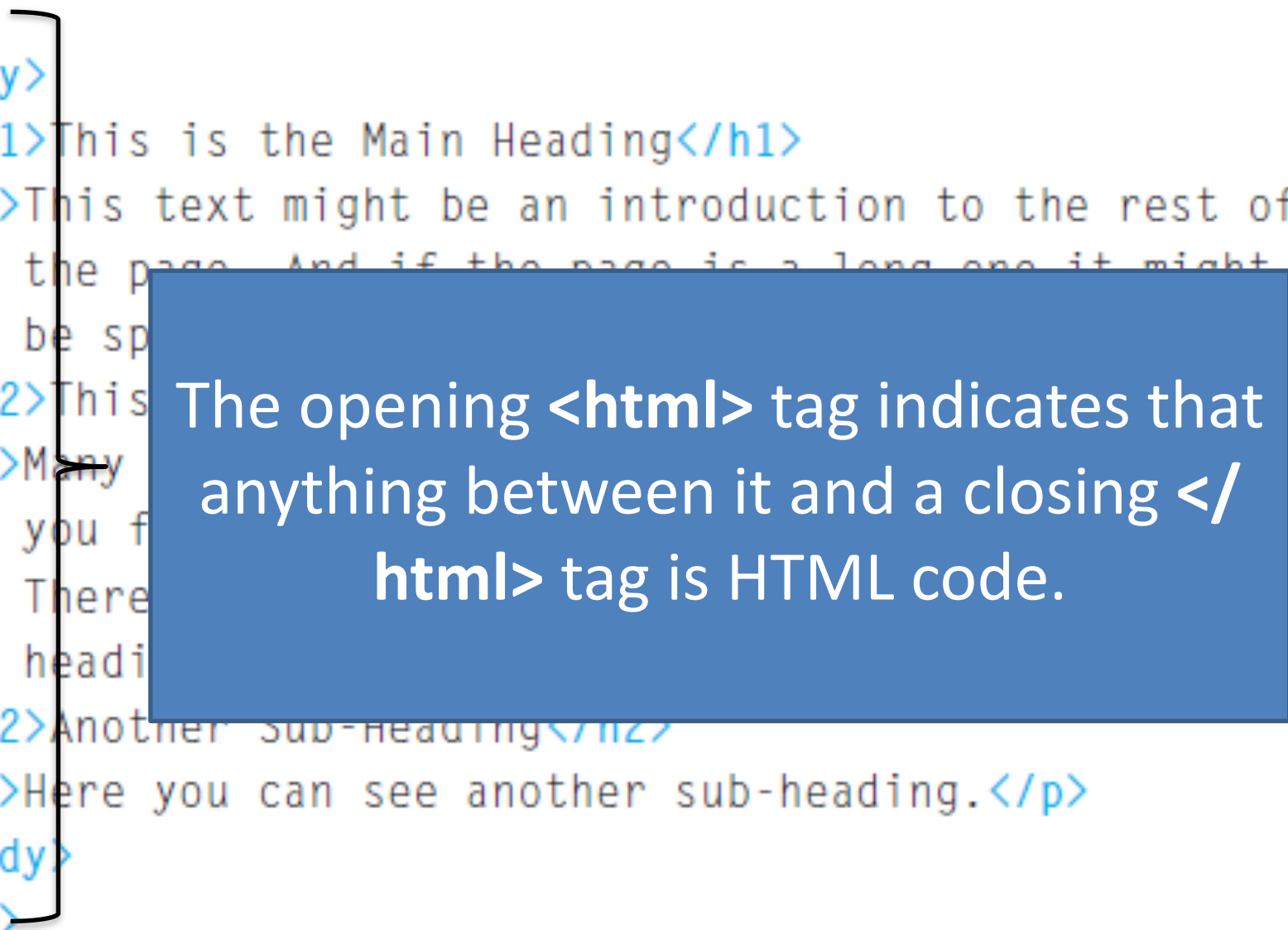
Many long articles have sub-headings so to help you follow the structure of what is being written. There may even be sub-sub-headings (or lower-level headings).

Another Sub-Heading

Here you can see another sub-heading.

HTML Structure

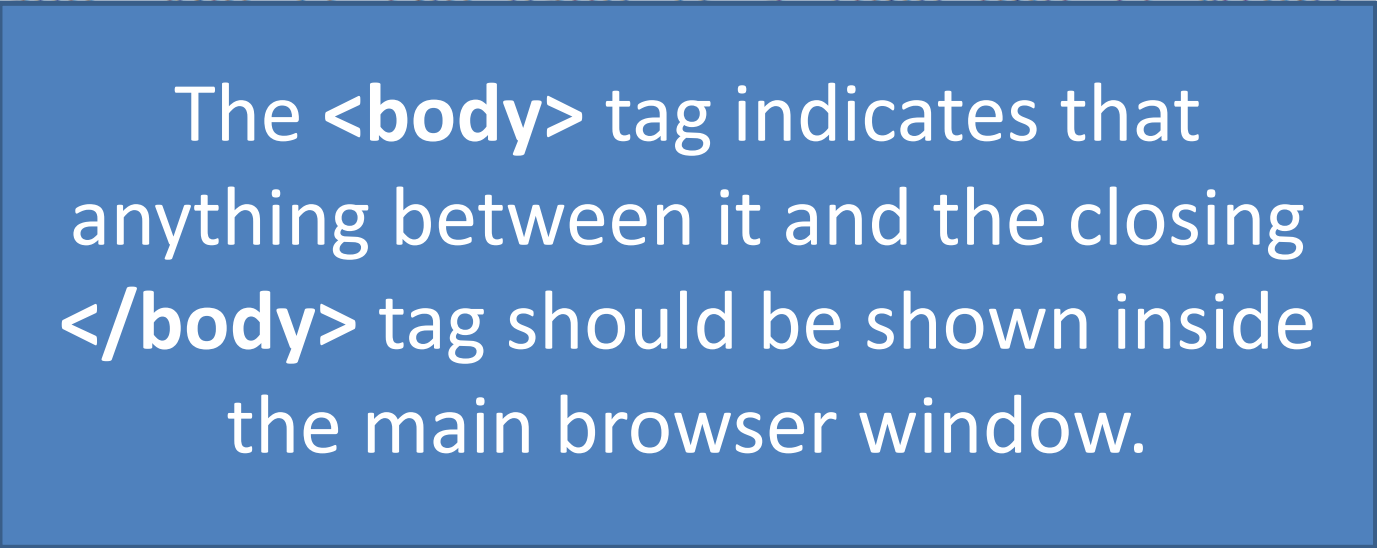
```
<html>
  <body>
    <h1>This is the Main Heading</h1>
    <p>This text might be an introduction to the rest of
      the page. And if the page is a long one it might
      be sp
    <h2>This
    <p>Many
      you f
      There
      headi
    <h2>Another sub-heading</h2>
    <p>Here you can see another sub-heading.</p>
  </body>
</html>
```



The opening `<html>` tag indicates that anything between it and a closing `</html>` tag is HTML code.

HTML Structure

```
<html>
  <body>
    <h1>This is the Main Heading</h1>
    <p>This text might be an introduction to the rest of
      the page. And if the page is a long one it might
      be sp
    <h2>This
    <p>Many
      you f
      There
      headi
    <h2>Another sub-heading</h2>
    <p>Here you can see another sub-heading.</p>
  </body>
</html>
```



The **<body>** tag indicates that anything between it and the closing **</body>** tag should be shown inside the main browser window.

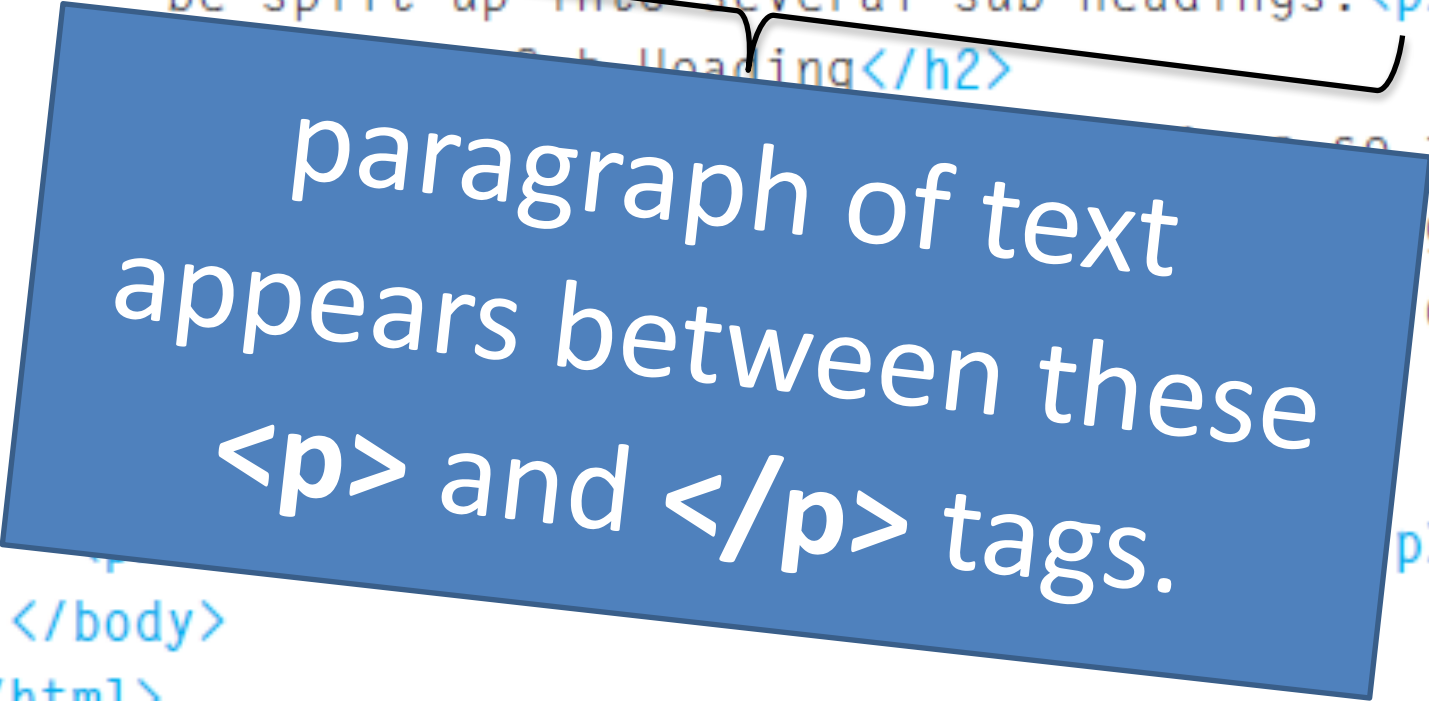
HTML Structure

```
<html>
  <body>
    <h1>This is the Main Heading</h1>
    <p>This text might be an introduction to the rest of
      the page. And if the page is a long one it might
      go on for a while. </p>
    <h2>Another sub-heading</h2>
    <p>Here you can see another sub-heading.</p>
  </body>
</html>
```

Words between `<h1>` and `</h1>` are a main heading.

HTML Structure

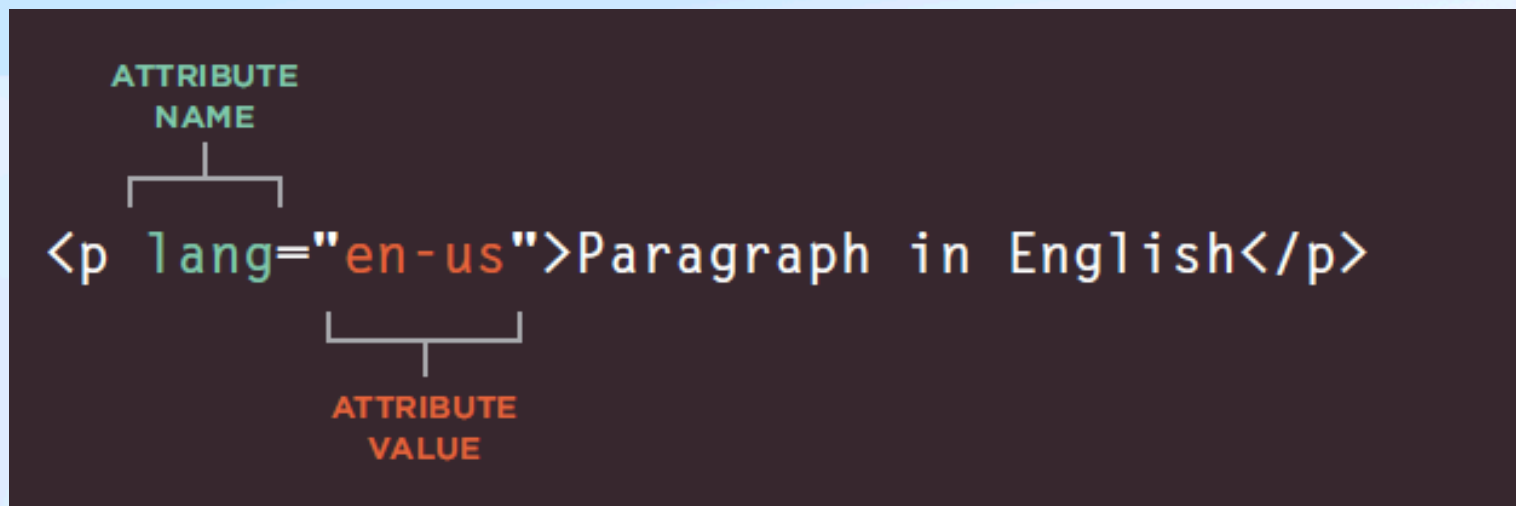
```
<html>
  <body>
    <h1>This is the Main Heading</h1>
    <p>This text might be an introduction to the rest of
      the page. And if the page is a long one it might
      be split up into several sub-headings.<p>
      Sub-Heading</h2>
      ... to help
      ... written.
      ... lower-level
      ... p>
  </body>
</html>
```



paragraph of text
appears between these
<p> and </p> tags.

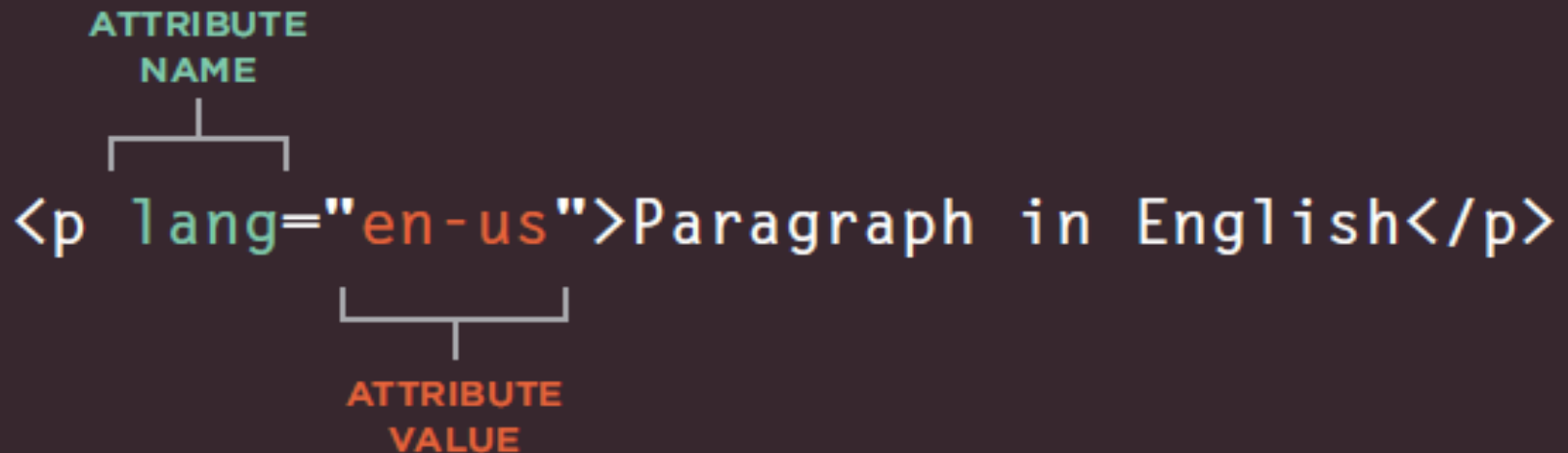
Attributes

- Attributes Tell Us More About Elements
 - Attributes provide additional information about the contents of an element. They appear on the opening tag of the element and are made up of two parts: a name and a value, separated by an equals sign.



HTML Tags

- Attributes Tell Us More About Elements



The **attribute name** indicates what kind of extra information you are supplying about the element's content. It should be written in lowercase.

The **value** is the information or setting for the attribute. It should be placed in double quotes. Different attributes can have different values.

Here an attribute called `lang` is used to indicate the language used in this element. The value of this attribute on this page specifies it is in US English.

Body & Title

```
<html>
  <head>
    <title>This is the Title of the Page</title>
  </head>
  <body>
    <h1>This is the body of the page</h1>
    <p>Anything within the body of a web page is
      displayed in the main browser window.</p>
  </body>
</html>
```

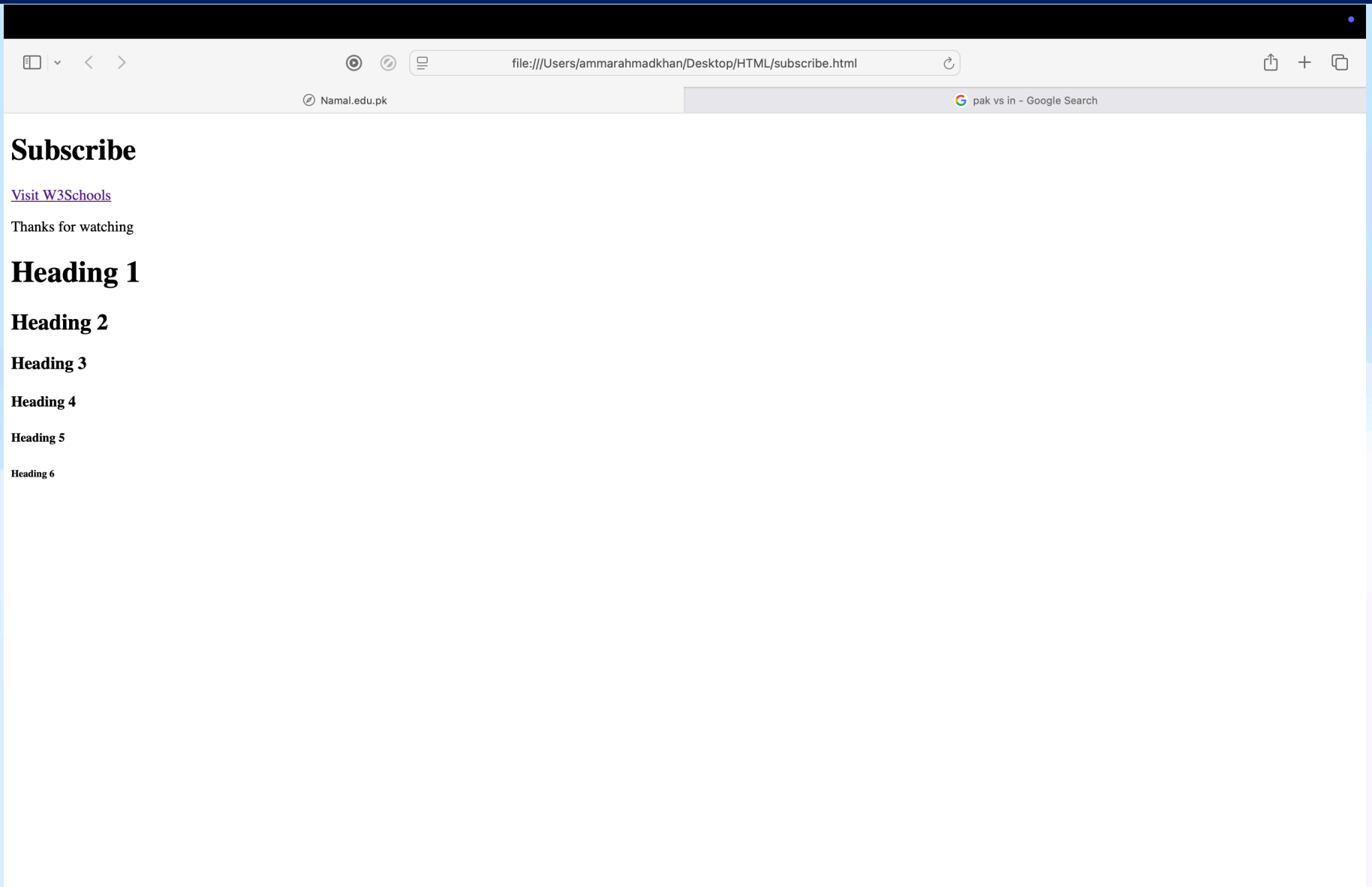
Everything inside body element is shown inside the main browser window.

Body & Title

```
<html>
  <head>
    <title>This is the Title of the Page</title>
  </head>
  <body>
    <h1>This is the Body of the Page</h1>
    <p>Anything within the body of a web page is
    displayed in the main browser window.</p>
  </body>
</html>
```

This contains information that are shown on top of the page

Body & Title



Headings

- HTML has six "levels" of headings:
 - **<h1>** is used for main headings
 - **<h2>** is used for subheadings
 - If there are further sections under the subheadings then the **<h3>** element is used, and so on...

Heading

- Example code

```
<html>
<head>
<title>This is the Title of the Page</title>
</head>
<body>
<h1>This is a Main Heading</h1>
<h2>This is a Level 2 Heading</h2>
<h3>This is a Level 3 Heading</h3>
<h4>This is a Level 4 Heading</h4>
<h5>This is a Level 5 Heading</h5>
<h6>This is a Level 6 Heading</h6>
</body>
</html>
```

This is main heading

This is level 2 heading

This is level 3 heading

This is level 4 heading

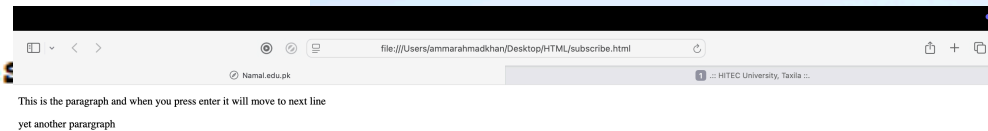
This is level 5 heading

This is level 6 heading

Paragraph

- A paragraph consists of one or more sentences that form a kind of dialogue.

```
<html>
<head>
<title>This is the Title of the Page</title>
</head>
<body>
<p>this is a paragraph and
when you press enter, does it moves
new line?</p>
<p>yet another paragraph</p>
</body>
</html>
```



Bold and Italic

- Bold
 - By enclosing words in the tags `<b>` and `</b>` we can make characters appear bold.
- Italic
 - By enclosing words in the tags `<i>` and `</i>` we can make characters appear italic.
- Example

```
<b>Does it appear bold</b>  
<i>and it must be italic</i>
```

Does it appear bold *and it must be italic*

Line Breaks & Horizontal Rules

- Line Break

- Just like `<p></p>` the line break `<br />` breaks line and starts new line
- Example

```
<p>The Earth<br />gets one hundred tons heavier  
every day<br />due to falling space dust.</p>
```

- Horizontal Rule

- Just like `<br />` it separates lines by placing a solid line in between paragraphs

```
<p>Venus is the only planet that rotates  
clockwise.</p>
```

```
<hr />
```

```
<p>Jupiter is bigger than all the other planets  
combined.</p>
```

Blockquote

- Blockquote
 - The `<blockquote>` tag specifies a section that is quoted from another source.
 - Browsers usually indent `<blockquote>` elements.

```
<!DOCTYPE html>
<html>
  <body>
    <h1>The blockquote element</h1>
    <p>Here is a quote from WWF's website:</p>
    <blockquote cite="http://www.worldwildlife.org/who/index.html">
      For 50 years, WWF has been protecting the future of nature....
    </blockquote>
  </body>
</html>
```

The blockquote element

Here is a quote from WWF's website:

For 50 years, WWF has been protecting the future of nature....

Abbreviation or Acronym

- Abbreviation or an acronym
 - Use an abbreviation or an acronym, when you like to add shortened versions of something else

```
<!DOCTYPE html>
<html>
<body>

<h1>The abbr element</h1>

<p>The <abbr title="World Health Organization">WHO</abbr> was founded in 1948.</p>

</body>
</html>
```

The abbr element

The WHO was founded in 1948.

Lists

- HTML provides us with three different types:
 - **Ordered lists**
 - are lists where each item in the list is numbered.
 - **Unordered lists**
 - are lists that begin with a bullet point (rather than characters that indicate order).
 - **Definition lists**
 - are made up of a set of terms along with the definitions for each of those terms.

List

- Ordered List

- `<ol>`

- The ordered list is created with the `<ol>` element.

- `<ul>`

- The unordered list is created with `<ul>` element.

- `<li>`

- Each item in the list is placed between an opening `<li>` tag
 - and a closing `</li>` tag. (The li stands for list item.)

- Browsers indent lists by default.

List

- Example

```
<!DOCTYPE html>
<html>
<body>

<h2>An Unordered HTML List</h2>

<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

<h2>An Ordered HTML List</h2>

<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

An Unordered HTML List

- Coffee
- Tea
- Milk

An Ordered HTML List

1. Coffee
2. Tea
3. Milk

List

- Definition List

- The definition list is created with the **<dl>** element and usually consists of a series of terms and their definitions.

```
<!DOCTYPE html>
<html>
<body>

<dl>
  <dt>Coffee</dt>
  <dd>Black hot drink</dd>
  <dt>Milk</dt>
  <dd>White cold drink</dd>
</dl>

</body>
</html>
```

Coffee	Black hot drink
Milk	White cold drink

Example HTML document

Scrambled Eggs

Eggs are one of my favourite foods. Here is a recipe for deliciously rich scrambled eggs.

Ingredients

- 2 eggs
- 1tbs butter
- 2tbs cream

Method

1. Melt butter in a frying pan over a medium heat
2. Gently mix the eggs and cream in a bowl
3. Once butter has melted add cream and eggs
4. Using a spatula fold the eggs from the edge of the pan to the center every 20 seconds (as if you are making an omelette)
5. When the eggs are still moist remove from the heat (it will continue to cook on the plate until served)

```
<html>
  <head>
    <title>Lists</title>
  </head>
  <body>
    <h1>Scrambled Eggs</h1>
    <p>Eggs are one of my favourite foods. Here is a
      recipe for deliciously rich scrambled eggs.</p>
    <h2>Ingredients</h2>
    <ul>
      <li>2 eggs</li>
      <li>1tbs butter</li>
      <li>2tbs cream</li>
    </ul>
    <h2>Method</h2>
    <ol>
      <li>Melt butter in a frying pan over a medium
        heat</li>
      <li>Gently mix the eggs and cream in a bowl</li>
      <li>Once butter has melted add cream and eggs</li>
      <li>Using a spatula fold the eggs from the edge of
        the pan to the center every 20 seconds (as if
        you are making an omelette)</li>
      <li>When the eggs are still moist remove from the
        heat (it will continue to cook on the plate
        until served)</li>
    </ol>
  </body>
</html>
```

Link

- Hyperlink
 - A **hyperlink** is a word, phrase, or image that you can click on to jump to a new document or a new section within the current document.
 - Example
 - `<a href="url">link text</a>`

Hyperlink

- Example

```
<!DOCTYPE html>
<html>
  <body>
    <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about-us.html">About</a></li>
      <li><a href="movies.html">Movies</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </body>
</html>
```

- [Home](#)
- [About](#)
- [Movies](#)
- [Contact](#)

Hyperlink

- Opening link in new window
 - Open link in new page by using *target="_blank"*

```
<!DOCTYPE html>
<html>
  <body>
    <p>
      <ul>
        <li><a href="index.html" target="_blank">Home</a></li>
        <li><a href="about-us.html">About</a></li>
        <li><a href="movies.html">Movies</a></li>
        <li><a href="contact.html">Contact</a></li>
      </ul>
    </p>
  </body>
</html>
```

Images

- There are three rules to remember when you are creating images.
 - Save images in the right format
 - Save images at the right size
 - Use the correct resolution

Figure & Figure Caption

- **<figure>** element to contain images
- The **** element is used to add images to a web page
- **<figcaption>** allow web page authors to add a caption to an image.

```
<figure>  
    
  <br />  
  <figcaption>Sea otters hold hands when they  
    sleep so they don't drift away from each  
    other.</figcaption>  
</figure>
```



Image – alt attribute

```

```



Tables

- A table represents information in a grid format.
- **Table**
 - The `<table>` element is used to create a table. The contents of the table are written out row by row.
- **Row**
 - start of each row using the opening `<tr>` tag and `</tr>` closing tag
- **Cell**
 - Each cell of a table is represented using a `<td>` element with `</td>` closing tag

Tables

```
<table>
  <tr>
    <td>15</td>
    <td>15</td>
    <td>30</td>
  </tr>
  <tr>
    <td>45</td>
    <td>60</td>
    <td>45</td>
  </tr>
  <tr>
    <td>60</td>
    <td>90</td>
    <td>90</td>
  </tr>
</table>
```

15	15	30
45	60	45
60	90	90

Tables

```
<table>
  <tr>
    <th></th>
    <th scope="col">Saturday</th>
    <th scope="col">Sunday</th>
  </tr>
  <tr>
    <th scope="row">Tickets sold:</th>
    <td>120</td>
    <td>135</td>
  </tr>
  <tr>
    <th scope="row">Total sales:</th>
    <td>$600</td>
    <td>$675</td>
  </tr>
</table>
```

	Saturday	Sunday
Tickets sold:	120	135
Total sales:	\$600	\$675

HTML Forms

- Whenever you want to collect information from visitors you will need a form, which lives inside a **<form>** element.
- Information from a form is sent in name/value pairs.
- Each form control is given a name, and the text the user types in or the values of the options they select are sent to the server.

HTML Form Structure

- **<form>**
 - This element should always carry the **action** attribute and will usually have a **method** and **id** attribute too.
- **Action**
 - Every **<form>** element requires an **action** attribute. Its **value** is the URL for the page on the server that will receive the information.
- **Method**
 - Forms can be sent using one of two methods: **get** or **post**.

HTML FORMS

```
<form action="http://www.example.com/login.php">
  <p>Username:
    <input type="text" name="username" size="15"
      maxlength="30" />
  </p>
  <p>Password:
    <input type="password" name="password" size="15"
      maxlength="30" />
  </p>
</form>
```

What does it means
(type="password")

Username:

Password:

HTML Page

- HTML page can be created on multiple text editor including:
 - Notepad
 - Notepad ++
 - Sublime
 - Visual Studio
 - Much more...

Markup Validation Service



Markup Validation Service

Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI

Validate by File Upload

Validate by Direct Input

Validate by File Upload

Upload a document for validation:

File:

Browse...

No file selected.

► [More Options](#)

Check

Note: file upload may not work with Internet Explorer on some versions of Windows XP Service Pack 2, see our [information page](#) on the W3C QA Website.

Markup Validation Service



Markup Validation Service

Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI

Validate by File Upload

Validate by Direct Input

Validate by URI

Validate a document online:

Address:

► [More Options](#)

Check

Markup Validation Service



Markup Validation Service

Check the markup (HTML, XHTML, ...) of Web documents

Validate by **URI**

Validate by **File Upload**

Validate by **Direct Input**

Validate by direct input

Enter the Markup to validate:

► [More Options](#)

Check

HTML Text formatting Tags

Tag	Description
<u></u>	Defines bold text
<u></u>	Defines emphasized text
<u><i></u>	Defines italic text
<u><small></u>	Defines smaller text
<u></u>	Defines important text
<u><sub></u>	Defines subscripted text
<u><sup></u>	Defines superscripted text
<u><ins></u>	Defines inserted text
<u></u>	Defines deleted text
<u><mark></u>	Defines marked/highlighted text

HTML Entities

Result	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	registered trademark	®	®

HTML Entities - Usage

- Some characters are reserved in HTML.
- If you use the less than (<) or greater than (>) signs in your text, the browser might mix them with tags.
- Character entities are used to display reserved characters in HTML.
- A character entity looks like this:
 - *&entity_name;* OR
 - *&#entity_number;*

HTML Entities – Math symbols

Result	Description	Entity Name	Entity Number
$\forall$	FOR ALL	∀	∀
∂	PARTIAL DIFFERENTIAL	∂	∂
$\exists$	THERE EXISTS	∃	∃
$\emptyset$	EMPTY SETS	∅	∅
∇	NABLA	∇	∇
$\in$	ELEMENT OF	∈	∈
$\notin$	NOT AN ELEMENT OF	∉	∉
$\ni$	CONTAINS AS MEMBER	∋	∋
$\prod$	N-ARY PRODUCT	∏	∏
$\sum$	N-ARY SUMMATION	∑	∑

More: http://www.w3schools.com/html/html_symbols.asp

iframe Tag

- An iframe is used to display a web page within a web page.
- Syntax
 - `<iframe src="URL"></iframe>`
- Iframe - Set Height and Width
 - `<iframe src="demo_iframe.htm" width="200" height="200"></iframe>`
- Iframe can be used to embed video in web page.
- Example: Embed following code in your HTML page
 - `<iframe src="https://www.youtube.com/embed/x7zE7Fu8Dmw" width="454" height="280" ></iframe>`

Example HTML Page

The Story in the Book

Chapter 1

Molly had been staring out of her window for about an hour now. On her desk, lying between the copies of *Nature*, *New Scientist*, and all the other scientific journals her work had appeared in, was a well thumbed copy of *On The Road*. It had been Molly's favorite book since college, and the longer she spent in these four walls the more she felt she needed to be free.

She had spent the last ten years in this room, sitting under a poster with an Oscar Wilde quote proclaiming that "Work is the refuge of people who have nothing better to do". Although many considered her pioneering work, unraveling the secrets of the llama DNA, to be an outstanding achievement, Molly *did* think she had something better to do.

Example HTML Document

```
<html>
  <head>
    <title>Text</title>
  </head>
  <body>
    <h1>The Story in the Book</h1>
    <h2>Chapter 1</h2>
    <p>Molly had been staring out of her window for about
      an hour now. On her desk, lying between the copies
      of <i>Nature</i>, <i>New Scientist</i>, and all
      the other scientific journals her work had
      appeared in, was a well thumbed copy of <cite>On
      The Road</cite>. It had been Molly's favorite book
      since college, and the longer she spent in these
      four walls the more she felt she needed to be
      free.</p>
    <p>She had spent the last ten years in this room,
      sitting under a poster with an Oscar Wilde quote
      proclaiming that <q>Work is the refuge of
      people who have nothing better to do</q>. Although
      many considered her pioneering work, unraveling
      the secrets of the llama <abbr
      title="Deoxyribonucleic acid">DNA</abbr>, to be an
      outstanding achievement, Molly <em>did</em> think
      she had something better to do.</p>
  </body>
</html>
```